General soln verified using case studies

$N = 1$ $\quad S = \ln E + \ln 2 + \ln \pi - \underbrace{\ln 0!}_{= \ln 1 = 0}$

$S = \ln 2 + \ln \pi$ ✓

$N = 2$ $\quad S = \ln E + 2\ln 2 + 2\ln \pi - \underbrace{\ln 1}_{= 0}$

$S = \ln E + 2\ln \pi + \ln 4$ ✓

$N = 3$ $\quad S = 2\ln E + \ln 8 + 3\ln \pi - \ln 2$

$S = 2\ln E + 3\ln \pi + \ln \frac{8}{2}$

$S = 2\ln E + 3\ln \pi + \ln 4.$